CS601 FINAL TERM SOLVED MCQs

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An unauthorized user is a network _____ issue.

- ► Performance
- ► Reliability
- ► **Security**
- ► All of the given

Question No: 2 (Marks: 1) - Please choose one

Which is not an element of protocol

- ► semantics
- ► timing
- ► **communication service module**

Question No: 3 (Marks: 1) - Please choose one

_____ is a multipoint topology.

- ► Ring
- ► Mesh

► Tree

► **Bus**

Question No: 4 (Marks: 1) - Please choose one

Unidirectional traffic movement is overcome by dual ring technology.

► **True**

► False

Question No: 5 (Marks: 1) - Please choose one

Physical layer define characteristics of interface between device and _____

► **transmission medium**

► another device

► another peer physical layer at other side

► modem

Question No: 6 (Marks: 1) - Please choose one

_____ layer deals with syntax and semantics of information exchange.

► **presentation**

► session

► application

► physical

Question No: 7 (Marks: 1) - Please choose one

To allow access to network resources is the function of _____

► **application layer**

► physical layer

► network layer

Question No: 8 (Marks: 1) - Please choose one

Time domain plot show changes in signal phase with respect to time.

► True

► **False**

Question No: 9 (Marks: 1) - Please choose one

In 8QAM each signal shift or one baud represents _____.

► 4 bits

► 2 bits

► 5 bits

► **3 bits**

Question No: 10 (Marks: 1) - Please choose one

Modulation of an analog signal can be accomplished through changing the _____ of the carrier signal.

► **amplitude**

► frequency

► phase

► all of the given

Question No: 11 (Marks: 1) - Please choose one

EIA 449 provides much better functionality than EIA _____

► **232**

► 223

► 262

► 222

Question No: 12 (Marks: 1) - Please choose one

Which of the following is an example of ITU-T modem standards:

► T-series

► X-series

► N-series

► **V-series**

Question No: 13 (Marks: 1) - Please choose one

Traditional modems are wide spread now to a data rate of _____.

► **56 Kbps**

► 72 Kbps

► 42 Kbps

► 96 Kbps

Question No: 14 (Marks: 1) - Please choose one

In case of uploading at the switching station, data is converted to digital signal using _____.

► TCP

► **PCM**

► ICP

► TDM

Question No: 15 (Marks: 1) - Please choose one

The RG number gives us information about _____.

► **Twisted pairs**

► Coaxial cables

► Optical fibers

► all of the given

Question No: 16 (Marks: 1) - Please choose one

The _____ is an association that sponsors the use of infrared waves.

► IrDA

► EIA

► **FCC**

► PUD

Question No: 17 (Marks: 1) - Please choose one

Optical fibers are defined by the ratio of the _____ of their core to the diameter of their cladding.

► **Diameter**

► Radius

► Length

► Width

Question No: 18 (Marks: 1) - Please choose one

The section of EM spectrum defined as Radio Communication is divided into _____ ranges called BANDS.

► **8**

► 10

► 5

► 6

Question No: 19 (Marks: 1) - Please choose one

Radio wave transmission utilizes _____ different types of propagation.

► Four

► Three

► Two

► **Five**

Question No: 20 (Marks: 1) - Please choose one

The VLF and LF bands use _____ propagation for communications.

► Ground

► Sky

► Line of sight

► **Space**

Question No: 21 (Marks: 1) - Please choose one

In _____ propagation, low-frequency radio waves hug the earth.

► Ground

► Sky

► Line of Sight

► **Space**

Question No: 22 (Marks: 1) - Please choose one

dB is _____ if a signal is amplified.

► Negative

► **Positive**

► Null

► Zero

Question No: 23 (Marks: 1) - Please choose one

Distortion occurs in a _____ signal.

► Rectified

► **Composite**

► Amplified

► none of the given

Question No: 24 (Marks: 1) - Please choose one

There are _____ basic categories of multiplexing.

► **3**

► 4

► 2

► 5

Question No: 25 (Marks: 1) - Please choose one

In bit _____, MUX adds extra bits to a device.

► **Stuffing**

► Adding

► Multiplication

► Exchanging

Question No: 26 (Marks: 1) - Please choose one

The local loop has _____ cable that connects the subscriber telephone to the nearest end office.

► Twisted-pair

► Coaxial

► Fiber-optic

► **None of the given**

Question No: 27 (Marks: 1) - Please choose one

FTTC stands for _____

► flexible to the curb

► **fiber to the curb**

► fiber to the cable

► fiber to the center

Question No: 28 (Marks: 1) - Please choose one

If the ASCII character G is sent and the character D is received, what type of error is this?

► **Single-bit**

► Multiple-bit

► Burst

► Recoverable

Question No: 29 (Marks: 1) - Please choose one

Which error detection method can detect a single-bit error?

► **Simple parity check**

► Two-dimensional parity check

► CRC

► All of the given

Question No: 30 (Marks: 1) - Please choose one

Flow control is needed to prevent _____

► Bit errors

► Overflow of the sender buffer

► **Overflow of the receiver buffer**

► Collision between sender and receiver

Question No: 31 (Marks: 1) - Please choose one

In data link layer, communication requires at least _____ devices working together

► 3

► 2

► 4

► 5

Question No: 32 (Marks: 1) - Please choose one

Data link control is composed of _____ important functions.

► 2

► 3

► 4

► 5

Question No: 33 (Marks: 1) - Please choose one

_____ coordinates the amount of data that can be sent before receiving acknowledgment

► flow control

► error control

► data control

Question No: 34 (Marks: 1) - Please choose one

Primary device uses _____ to receive transmission from the secondary devices.

► ACK

► ENQ

► **POLL**

Question No: 35 (Marks: 1) - Please choose one

In a Go-Back-N ARQ, if the window size is 63, what is the range of sequence number?

► **0 to 63**

► 0 to 64

► 1 to 63

► 1 to 64

Question No: 36 (Marks: 1) - Please choose one

Data link protocols can be divided into _____ sub-groups.

► **two**

► three

► four

► five

Question No: 37 (Marks: 1) - Please choose one

XMODEM is a _____ protocol designed for telephone-line communication b/w PCs.

► **file transfer**

► hardware

► software

► application exchange

Question No: 38 (Marks: 1) - Please choose one

YMODEM uses ITU-T CRC-_____ for Error Checking

► **16**

► 32

► 8

► 4

Question No: 39 (Marks: 1) - Please choose one

In Y-MODEM Multiple files can be sent simultaneously

► **True**

► False

Question No: 40 (Marks: 1) - Please choose one

HDLC is an acronym for _____.

► High-duplex line communication

► **High-level data link control**

► Half-duplex digital link combination

► Host double-level circuit

Question No: 41 (Marks: 1) - Please choose one

What is present in all HDLC control fields?

► P/F bit

► N(R)

► N(S)

► Code bits

Question No: 42 (Marks: 1) - Please choose one

Which of the following sublyer, resolves the contention for the shared media

► **MAC**

► LLC

► Phyical

Question No: 43 (Marks: 1) - Please choose one

Ethernet LANs can support data rates between _____

► **1 and 100 Mbps**

► 1 and 200 Mbps

► 1 and 500 Mbps

► 1 and 100 Gbps

Question No: 44 (Marks: 1) - Please choose one

In FDDI, Token Passing is used as Access method.

- ► **True**
- ► False

Question No: 45 (Marks: 1) - Please choose one

Bridges can divide a large _____ into smaller segments

- ► **Network**
- ► Packet
- ► Frame
- ► Address

Question No: 46 (Marks: 1) - Please choose one

Like VRC, LRC and CRC, Checksum is also based on _____.

- ► **Redundency**
- ► Decimal Division
- ► Encryption
- ► Encoding

Question No: 47 (Marks: 1) - Please choose one

Check sum method is used for _____ layers.

► Physical

► Application

► Transport

► Datalink

Question No: 48 (Marks: 1) - Please choose one

Which of the following _____ uses a series of filters to decompose multiplexed signal into its constituent signals.

► Bridge

► MUX

► **DEMUX**

► Switch

Question No: 49 (Marks: 1) - Please choose one

We need _____ to decompose a composite signal into its components.

► fourier transform

► nyquist theorem

► shannon capacity

Question No: 50 (Marks: 1) - Please choose one

Data from computer is in _____ form and the local loop handles _____ signals.

► Analog; analog

► Analog; digital

► Digital; digital

► **Digital; analog**

Question No: 51 (Marks: 2)

What is the formula to calculate the number of redundancy bits required to correct a bit error in a given number of data bits? [2]

Messages(frames) consist of m data (message) bits, yielding an n=(m+r)-bit codeword.

Question No: 52 (Marks: 2)

What is R G rating of coaxial cable?
Different coaxial cable designs are categorized by their Radio government (RG) ratings
Each cable defined by RG rating is adapted for a specialized function:
RG-8
- Used in Thick Ethernet

RG-9
- Used in Thick Ethernet

RG-11
- Used in Thick Ethernet

RG-58
- Used in Thin Ethernet

RG-59
- Used for TV

Question No: 53 (Marks: 2)

What are the advantages of thin ethernet?
The advantages of thin Ethernet are :
- reduced cost and

- ease of installation

Because the cable is lighter weight and more flexible than that used in Thicknet

Question No: 54 (Marks: 3)

What is the difference between a unicast, multicast, and broadcast address? [3]

Three methods can be used to transmit packets over a network: unicast, multicast, and broadcast.

Unicast involves communication between a single sender and a single receiver. This is a type of point-to-point transmission; since the packet is transmitted to one destination at a time.

Multicast is used to send packets to a group of addresses, represented by a "group address." In this case, packets are transmitted from a single sender to multiple receivers. Since the same data packet can be sent to multiple nodes by sending just one copy of the data, the load of the sender and the overall load of the network are both reduced.

Broadcast involves sending packets to all nodes on a network simultaneously. This type of transmission is used to establish communication with another host, and for DHCP type methods of assigning IP addresses.

Question No: 55 (Marks: 3)

T lines are designed for Digital data how they can be used for Analog Transmission ?

T Lines are digital lines designed for digital data however; they can also be used for analog transmission (Telephone connections). Analog signals are first sampled and the Time Multiplexed.

Question No: 56 (Marks: 3)

What are the three types of Guided Media?

Guided Media, are those media that provide a conduit from one device to another. Three types are

1. Twisted pair cable
2. Coaxial cable
3. Fiber-optic Cable

Question No: 57 (Marks: 5)

Why do we need Inverse Multiplexing? [5]
Data & Video can be broken into smaller portions using Inverse Multiplexing and TX. An inverse multiplexer (often abbreviated to "inverse mux" or "imux") allows a data stream to be broken into multiple lower data rate communication links. An inverse multiplexer differs from a demultiplexer in that each of the low rate links coming from it is related to the others and they all work together to carry their respective parts of the same higher rate data stream. By contrast, the output streams from a demultiplexer may be completely independent from each other and the demultiplexer does not have to understand them in any way.

This is the opposite of a multiplexer which creates one high speed link from multiple low speed ones.

It can lease a 1.544 Mbps line from a common carrier and only use it fully for sometime
Or it can lease several separate channels of lower data rates

Voice can be sent over any of these channels

Question No: 58 (Marks: 5)

Describe method of checksum briefly?
The sender subdivides data units into equal segments of ‘n’ bits(16 bits).These segments are added together using one’s complement. The total (sum) is then complemented and appended to the end of the original data unit as redundancy bits called CHECKSUM. The extended data unit is transmitted across the network. The receiver subdivides data unit as above and adds all segments together and complement the result. If the intended data unit is intact, total value found by adding the data segments and the checksum field should be zero. If the result is not zero, the packet contains an error & the receiver rejects it

Question No: 59 (Marks: 10)

Explain Asynchronous Time Division Multiplexing in detail? Also discuss its advantages over synchronous TDM?
Asynchronous time-division multiplexing (ATDM) is a method of sending

information that resembles normal TDM, except that time slots are allocated as needed dynamically rather than preassigned to specific transmitters. ATDM is more intelligent and has better bandwidth efficiency than TDM.

Time-division multiplexing (TDM) is a type of digital or (rarely) analog multiplexing in which two or more signals or bit streams are transferred apparently simultaneously as sub-channels in one communication channel, but are physically taking turns on the channel. The time domain is divided into several recurrent timeslots of fixed length, one for each sub-channel. A sample byte or data block of sub-channel 1 is transmitted during timeslot 1, sub-channel 2 during timeslot 2, etc. One TDM frame consists of one timeslot per sub-channel. After the last sub-channel the cycle starts all over again with a new frame, starting with the second sample, byte or data block from sub-channel 1, etc.

asynchronous time-division multiplexing comprising receive circuits (CRI/i) supplying cells received via input links, transmit circuits (CTI/j) transmitting retransmitted cells on output links, a buffer memory (MT) storing the received cells and delivering the cells to be retransmitted and a buffer memory addressing device (SMT) including a write address source (SAE) and a read address source (fsl/j).

The switching unit further comprises a write disabling circuit (pi) conditioned by a signal (adl) derived from the content of at least one received cell or a signal (tle) derived from the absence of any received cell and supplying a disabling signal (spi) and the address source includes a disabling device (pac, pal) influenced by the disabling signal (spi) so that no memory location is then occupied in the buffer memory (MT).

Advantages asynchronous TDM:

In asynchronous TDM, the timeslots are not fixed. They are assigned dynamically as needed.

In order to reduce the communications costs in time-sharing systems and multicomputer communication systems, multiplexing techniques have been introduced to increase channel utilization. A commonly used technique is Synchronous Time Division Multiplexing (STDM). In Synchronous Time Division Multiplexing, for example, consider the transmission of messages from terminals to computer, each terminal is assigned a fixed time duration. After one user's time duration has elapsed, the channel is switched to another user. With synchronous operation, buffering is limited to one character per user line, and addressing is usually not required. The STDM technique, however, has certain disadvantages. As shown in Figure 1, it is inefficient in capacity and cost to permanently assign a segment of bandwidth that is utilized only for a portion of the time. A more flexible system that efficiently uses the transmission facility on an "instantaneous time-shared" basis could be used instead. The objective would

be to switch from one user to another user whenever the one user is idle, and to asynchronously time multiplex the data. With such an arrangement, each user would be granted access to the channel only when he has a message to transmit. This is known as an Asynchronous Time Division Multiplexing System (ATDM). A segment of a typical ATDM data stream is shown in Figure 2. The crucial attributes of such a multiplexing technique are:

1. An address is required for each transmitted message, and

2. Buffering is required to handle the random message arrivals.

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